

Nonlinear Disk Obstructions for the Bidisc Carathéodory-Fejér Problem

fa_banach_001 lane 5

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Status

This is a partial result for arXiv:1508.07199, Rajeev Gupta, *The Carathéodory-Fejér Interpolation Problems and the von-Neumann Inequality* [1]. It gives additional necessary conditions for the degree-two bidisc Carathéodory-Fejér problem and completely solves a two-term subfamily. It does not classify all pairs p_1, p_2 in the CF class.

Source Question

In Section 4.4 of the source, after proving that the necessary condition $\|\mathcal{T}(M_{p_1}, M_{p_2})\| \leq 1$ is not sufficient in general, Gupta asks what additional properties on p_1, p_2 ensure membership in the CF class.

As in Subcase 1, here also $\|J\| = 1$ can be checked easily. $\square$

Remark 4.20. In Problem (4.10), If either $p_1 \equiv 0$ or $p_2 \equiv 0$, and $\|\mathcal{T}(M_{p_1}, M_{p_2})\| \leq 1$, then $\|P\| \leq 1$ and hence f in the Theorem (4.16) can be taken to be P itself.

Having verified that the necessary condition $\|\mathcal{T}(M_{p_1}, M_{p_2})\| \leq 1$ is also sufficient for P to be in the CF class in several cases, we expected it to be sufficient in general. But unfortunately this is not the case. We give an example of a polynomial P for which $\|\mathcal{T}(M_{p_1}, M_{p_2})\| \leq 1$ but P is not in the CF class.

An Example: Let $p_1(z) = 1/\sqrt{2}$ and $p_2(z) = z^2/2$. We show that $\mathcal{T}(M_{p_1}, M_{p_2})$ is not even 3-polynomially extendible.

It can easily be seen that $\|\mathcal{T}(M_{p_1}, M_{p_2})\| \leq 1$. Now suppose there exists a polynomial p_3 of degree at most 3 such that $\|\mathcal{T}(M_{p_1}, M_{p_2}, M_{p_3})\| \leq 1$. Then Parrott's theorem guarantees the existence of a contraction $V \in \mathcal{B}(L^2(\mathbb{T}))$ such that

$$M_{p_3} = \left(I - M_{|p_1|^2} - M_{p_2} \left(I - M_{|p_1|^2} \right)^{-1} M_{p_2}^* \right) V - M_{p_2} \left(I - M_{|p_1|^2} \right)^{-\frac{1}{2}} M_{p_1}^* \left(I - M_{|p_1|^2} \right)^{-\frac{1}{2}} M_{p_2}.$$

As we have $(1 - |p_1|^2)^2 - |p_2|^2 \equiv 0$, therefore operator in the first bracket is zero and hence

$$p_3 = \frac{-p_2^2 \overline{p_1}}{1 - |p_1|^2} = \sqrt{2} z^4.$$

Thus p_3 is a polynomial of degree more than 3 which is a contradiction. Hence $\mathcal{T}(M_{p_1}, M_{p_2})$ is not even 3-polynomially extendible.

We close this section with an open question: What are the properties we must impose on the polynomials p_1 and p_2 in addition to the requirement $\|\mathcal{T}(M_{p_1}, M_{p_2})\| \leq 1$ to ensure that P is in the CF class?

For

$$p(z_1, z_2) = a_{10}z_1 + a_{01}z_2 + a_{20}z_1^2 + a_{11}z_1z_2 + a_{02}z_2^2,$$

the source uses

$$p_1(\lambda) = a_{10} + a_{01}\lambda, \quad p_2(\lambda) = a_{20} + a_{11}\lambda + a_{02}\lambda^2.$$

The source's explicit necessary condition is

$$|p_2(\lambda)| \leq 1 - |p_1(\lambda)|^2 \quad (|\lambda| = 1).$$

Proof Intuition

The source's condition comes from testing an extension on the linear diagonal disks $(z_1, z_2) = (t, \lambda t)$. But a bidisc Schur function can also be tested on nonlinear analytic disks. The disks

$$(z_1, z_2) = (\alpha t^2, \beta t) \quad \text{and} \quad (z_1, z_2) = (\alpha t, \beta t^2)$$

mix one linear coefficient with one pure quadratic coefficient in the second Taylor coefficient of a one-variable Schur function. The tail q of a CF extension has no total degree ≤ 2 , so it cannot change those first two one-variable coefficients. The elementary one-variable Schur coefficient inequality therefore gives extra constraints.

One-Variable Input

Lemma 1 (Schur coefficient bound). *Let $F(t) = c_1t + c_2t^2 + O(t^3)$ be holomorphic from $\mathbb{D}$ to $\mathbb{D}$. Then*

$$|c_2| \leq 1 - |c_1|^2.$$

Proof. Since $F(0) = 0$, Schwarz's lemma gives $F(t) = tG(t)$ with G in the Schur class. Then $G(0) = c_1$ and $G'(0) = c_2$. The Schwarz-Pick lemma applied to G at 0 gives $|G'(0)| \leq 1 - |G(0)|^2$. $\square$

Nonlinear Disk Obstructions

Theorem 1. *Suppose there is a holomorphic $q : \mathbb{D}^2 \rightarrow \mathbb{C}$ such that*

$$q^{(k)}(0) = 0 \quad (|k| \leq 2), \quad \|p + q\|_{\mathbb{D}^2, \infty} \leq 1.$$

Then the coefficients of p satisfy

$$|a_{10}| + |a_{02}| \leq 1 - |a_{01}|^2 \tag{1}$$

and

$$|a_{01}| + |a_{20}| \leq 1 - |a_{10}|^2. \tag{2}$$

Proof. Let $f = p + q$. Fix unimodular α, β and set

$$F_{\alpha, \beta}(t) := f(\alpha t^2, \beta t).$$

This maps $\mathbb{D}$ into $\mathbb{D}$. Since every monomial of q has total degree at least 3, substituting $z_1 = \alpha t^2$, $z_2 = \beta t$ produces only powers $t^3, t^4, \dots$ from q . Therefore

$$F_{\alpha, \beta}(t) = a_{01}\beta t + (a_{10}\alpha + a_{02}\beta^2)t^2 + O(t^3).$$

Lemma 1 gives

$$|a_{10}\alpha + a_{02}\beta^2| \leq 1 - |a_{01}|^2.$$

Choosing α and β so that the two summands have the same argument gives (1).

The proof of (2) is symmetric. For unimodular α, β , set

$$G_{\alpha, \beta}(t) := f(\alpha t, \beta t^2).$$

Again the tail q contributes no terms of order 1 or 2, and

$$G_{\alpha, \beta}(t) = a_{10}\alpha t + (a_{01}\beta + a_{20}\alpha^2)t^2 + O(t^3).$$

Lemma 1 gives

$$|a_{01}\beta + a_{20}\alpha^2| \leq 1 - |a_{10}|^2,$$

and maximizing over phases gives (2). $\square$

A Sharp Two-Term Classification

Corollary 1. *For $a, b \in \mathbb{C}$, the polynomial*

$$p_{a,b}(z_1, z_2) = az_1 + bz_2^2$$

has a degree-two bidisc CF extension if and only if

$$|a| + |b| \leq 1.$$

Proof. Necessity is Theorem 1 with

$$a_{10} = a, \quad a_{02} = b, \quad a_{01} = a_{20} = a_{11} = 0.$$

For sufficiency, take $q = 0$. If $|a| + |b| \leq 1$, then for $(z_1, z_2) \in \mathbb{D}^2$,

$$|az_1 + bz_2^2| \leq |a||z_1| + |b||z_2|^2 \leq |a| + |b| \leq 1.$$

Thus $p_{a,b}$ itself is already a Schur-class extension. □

Corollary 2 (Counterexample family to sufficiency of the source condition). *Let $0 < |a| < 1$ and*

$$1 - |a| < |b| \leq 1 - |a|^2.$$

Then $p_{a,b}(z_1, z_2) = az_1 + bz_2^2$ satisfies the source necessary condition $|p_2| + |p_1|^2 \leq 1$, but has no degree-two bidisc CF extension.

Proof. Here $p_1(\lambda) = a$ and $p_2(\lambda) = b\lambda^2$, so

$$|p_2(\lambda)| + |p_1(\lambda)|^2 = |b| + |a|^2 \leq 1.$$

But $|a| + |b| > 1$, so Corollary 1 rules out a CF extension. □

Verification Notes

The proof is analytic and uses only the Schwarz-Pick coefficient estimate. The script `code/verify_disk_obstructions.py` performs a small numerical sanity check: it samples random coefficients and phases to verify that the maximized second coefficient on the nonlinear disks agrees with $|a_{10}| + |a_{02}|$ and $|a_{01}| + |a_{20}|$, and it checks the two-term classification boundary for sample values including the source example. This code is not part of the proof.

The source PDF was checked at PDF page 73, printed page 61, where the example $p_1 = 1/\sqrt{2}$, $p_2 = z^2/2$ and the open question appear.

Limitations and Novelty Check

This result is a partial answer to the source question. It supplies additional necessary conditions and a sharp classification of one two-term family, but it does not give necessary and sufficient conditions for arbitrary quadratic bidisc data.

Bounded novelty checks were performed in the local run indexes for ‘1508.07199’, ‘CF class’, ‘completely polynomially extendible’, and the source example $p_1 = 1/\sqrt{2}$, $p_2 = z^2/2$. Web searches on 2026-07-19 found the source thesis and the later Gupta-Misra *Studia Mathematica* paper, but no separate record of this nonlinear-disk obstruction or the exact $az_1 + bz_2^2$ classification. The novelty confidence is therefore moderate, not definitive.

Human Review Recommendation

Review Theorem 1. The key point is that the vanishing condition on q by total degree really prevents q from altering the first two coefficients after the nonlinear disk substitutions. Once that is checked, the rest is the one-variable Schwarz-Pick coefficient bound and phase maximization.

References

- [1] R. Gupta, *The Carathéodory-Fejér Interpolation Problems and the von-Neumann Inequality*, arXiv:1508.07199, 2015.
- [2] R. Gupta and G. Misra, *The Carathéodory-Fejér interpolation on the polydisc*, Studia Math. 254 (2020), 265–293.